

Sedjro Salomon Hotegni

MACHINE LEARNING ENGINEER | SCIENTIFIC ML & RESEARCH SOFTWARE

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PROFESSIONAL SUMMARY

Final-year PhD candidate in Computer Science and applied machine learning researcher with 3+ years of experience designing, implementing, and evaluating models in Python and PyTorch. Published at ICLR and ICML, with a track record of developing original algorithms, rigorous experimental benchmarks, and reusable research software. Experience spans generative modelling, representation learning, ranking and personalization, transformer-based text embeddings, and ML deployment. Interested in translating advanced deep learning methods into reliable solutions for complex datasets at the interface of computational and experimental research.

CORE SKILLS

Applied ML: PyTorch, TensorFlow, deep learning, diffusion models, flow matching, representation learning, adversarial robustness, multi-objective optimization, Bayesian optimization
Ranking & personalization: Recommender systems, implicit-feedback modelling, semantic retrieval, cold-start strategies, diversity-aware reranking, Recall@K, NDCG@K, MRR
ML systems & research engineering: FAISS, PostgreSQL/pgvector, FastAPI, Docker, GitHub Actions, pytest, Linux, Conda, Slurm/HPC, PyPI packaging, artifact versioning, latency benchmarking, experimental design, reproducible workflows, technical documentation
Programming & data: Python (advanced), NumPy, pandas, SciPy, SQL, R, C/C++, Java, Git
Languages: English (fluent), French (native), German (intermediate)

SELECTED PROJECT

SemanticCart: Hybrid Product Recommendation System

Independent machine learning engineering project

2026
Python | PyTorch | OpenAI | FAISS

- Built an end-to-end recommender over **814,586** chronological Amazon interactions, **94,762** users, and **25,612** products, combining implicit ALS, 512-dimensional OpenAI embeddings, recent-session intent, and diversity-aware MMR reranking.
- Improved frozen-test **Recall@10** by **7.2%**, **NDCG@10** by **9.0%**, **MRR@10** by **10.3%**, and catalogue coverage by 25.4% over ALS; selected all hyperparameters on validation before one-time test evaluation.
- Implemented cold-start and similar-item retrieval with FAISS HNSW (**0.416 ms p95**) and validated pgvector HNSW at 99.96% mean Top-10 overlap with FAISS.
- Packaged a versioned 331.77 MiB model bundle behind FastAPI, PostgreSQL, and Docker; measured **49.783 ms end-to-end p95** over 500 sequential loopback requests and enforced quality with 248 tests and four CI gates.

RESEARCH & MACHINE LEARNING ENGINEERING EXPERIENCE

TU Dortmund University - Safe Autonomous Systems Group

Research Scientist (Multi-Objective Machine Learning)

Oct. 2024 - Present
Dortmund, Germany

- Designed and implemented **SPREAD** (ICLR 2026), a PyTorch/DDPM framework that learns and refines Pareto fronts for multi-objective optimization.
- Built evaluation workflows across 11 synthetic and real-world tasks; ranked 1st on 9/11 tasks for hypervolume and 8/11 for diversity in online settings, and achieved the best average ranking across offline benchmarks.
- Packaged the method as **moospread** and released it on PyPI for reusable research workflows.
- Developed and benchmarked **MOREL** (ICANN 2025), a PyTorch representation-learning framework that improved AutoAttack accuracy by up to 2 percentage points on CIFAR-100 and at least 1 point on Tiny-ImageNet; released **advmorel** on PyPI.

Paderborn University - Data Science for Engineering Group

Research Scientist (Multi-Objective Machine Learning)

Dec. 2022 - Sep. 2024
Paderborn, Germany

- Designed a PyTorch training workflow to jointly optimize neural-network sparsity and predictive performance through the **MDMTN** multi-task architecture (IJCNN 2024), covering method development, implementation, and empirical evaluation.

Fatima Fellowship

Research Intern

May 2022 - Nov. 2022
Remote

- Developed approximation algorithms for fair range clustering with group interval constraints; obtained constant-factor guarantees through structured linear-programming relaxations and rounding (ICML 2023).
- Collaborated remotely with an international research team and translated complex theoretical findings into clear technical arguments and reproducible results.

EDUCATION

TU Dortmund University (previously Paderborn University, Dec. 2022 - Sep. 2024)

Ph.D. in Computer Science (expected Nov. 2026)

2022 - Present
Dortmund, Germany

Dissertation: Multi-Objective Optimization for Efficient, Robust, and Scalable Machine Learning

African Institute for Mathematical Sciences (AIMS)

M.Sc. in Mathematical Sciences (Data Science stream), Distinction

2021 - 2022
Kigali, Rwanda

Full scholarship.

University of Abomey-Calavi (UAC)

B.Sc. in Mathematics and Applications (Mathematics and Computing stream)

2017 - 2020
Abomey-Calavi, Benin

Full scholarship.

SELECTED PUBLICATIONS

- ICLR 2026.** S. S. Hotegni and S. Peitz. SPREAD: Sampling-based Pareto Front Refinement via Efficient Adaptive Diffusion.
- ICANN 2025.** S. S. Hotegni and S. Peitz. Enhancing Adversarial Robustness Through Multi-Objective Representation Learning.
- IJCNN 2024.** S. S. Hotegni, M. Berkemeier, and S. Peitz. Multi-Objective Optimization for Sparse Deep Multi-Task Learning.
- ICML 2023.** S. S. Hotegni, S. Mahabadi, and A. Vakilian. Approximation Algorithms for Fair Range Clustering.

SELECTED OPEN-SOURCE SOFTWARE

- moospread (PyPI)** - Reusable PyTorch package implementing SPREAD for sampling-based Pareto front learning.
- advmorel (PyPI)** - PyTorch package for multi-objective representation learning and adversarial robustness.

TECHNICAL COMMUNICATION & TEACHING

TU Dortmund University

Teaching Assistant

Oct. 2025 - Present
Dortmund, Germany

Lead exercise sessions and assess assignments for *Mathematics for Computer Science* and *Software Engineering*, explaining technical concepts to students with varied backgrounds.

EURO 2025

Conference talk

Jun. 2025
Leeds, UK

A Diffusion-Driven Generative Framework for Pareto Front Learning

DALI 2025

Research poster

Apr. 2025
Sorrento, Italy

Enhancing Adversarial Robustness through Multi-Objective Representation Learning

EUCCO 2023

Conference talk

Sep. 2023
Heidelberg, Germany

Multi-Objective Optimization for Deep Neural Network Training

SELECTED PROFESSIONAL DEVELOPMENT

- Google Data Analytics Professional Certificate** - Google (Coursera), 2021. Data preparation, processing, analysis, and visualization.
- Algorithmic Toolbox** - UC San Diego (Coursera), 2026. Algorithm design using divide-and-conquer, greedy methods, and dynamic programming.
- 3D Reconstruction - Multiple Viewpoints** - Columbia University (Coursera), 2026. Camera geometry, calibration, stereo vision, and multi-view reconstruction.